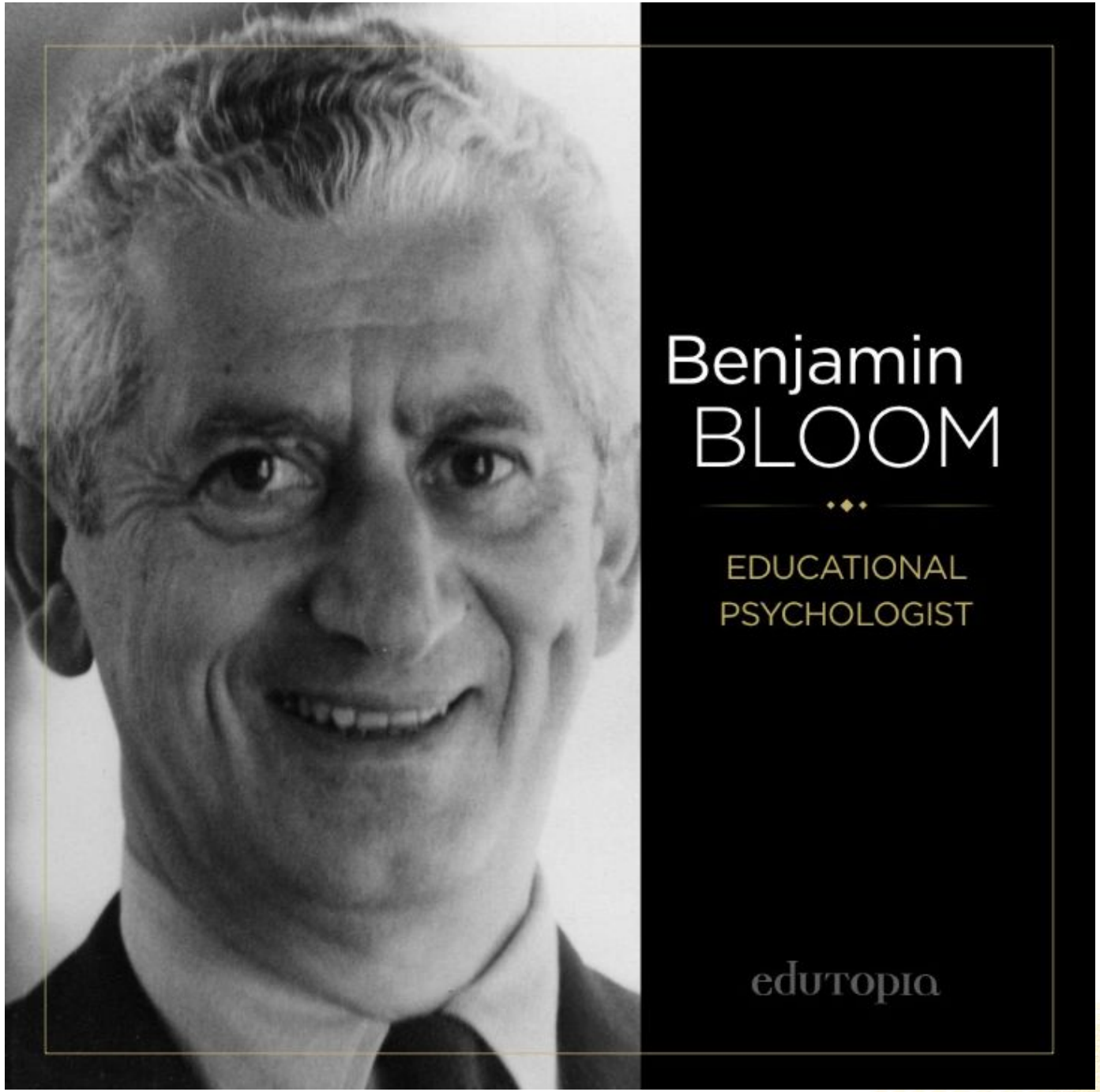
FATE Foundation UCL AIMS African Institute for Mathematical Sciences SOUTH AFRICA

PEDAGOGICAL CONCEPTS

Kenya - workshop

AI Research Foundations

with support from Google.org


Benjamin
BLOOM
EDUCATIONAL
PSYCHOLOGIST
edutopia

Learning Objective

DEFINITION A clear, specific statement describing what a learner is expected to be able to do by the end of a lesson, course, or activity : always starting with an action verb.

Examples:

- By the end of this lesson, students will be able to understand tokenization and compare different tokenization methods.
- By the end of this course, students will be able to explain how the BPE algorithm builds a vocabulary step by step.

Three Domains of Bloom's Taxonomy

COGNITIVE

How learners process knowledge

IN SCOPE TODAY

AFFECTIVE

How learners engage with values, attitudes and ethics

IN SCOPE TODAY

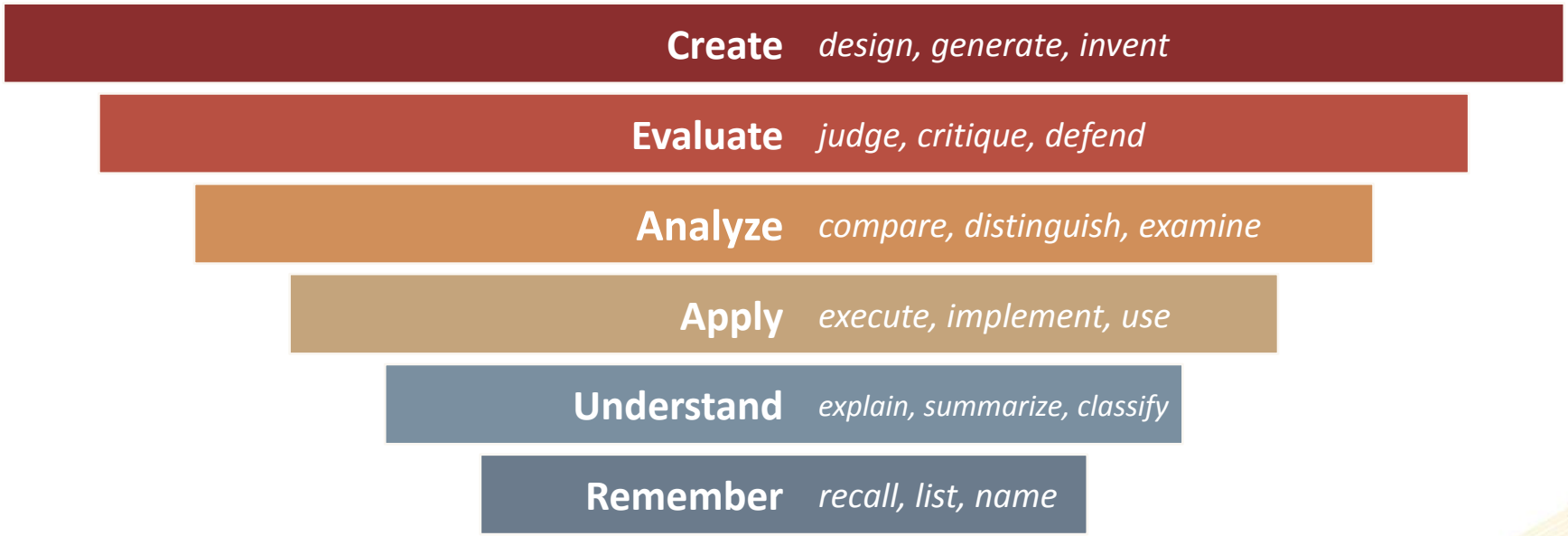
PSYCHOMOTOR

How learners develop physical skills through practice and hands-on experience

out of scope

The Cognitive Domain

DEFINITION Describes how learners develop knowledge and skills, from remembering facts to creating original work.



Cognitive Domain

REMEMBER · L1

The ability to remember and recall previously learned facts, concepts, or skills.

Example

List the five steps of the BPE algorithm used to create subword tokens from text data.

List | Recall | Name

UNDERSTAND · L2

The ability to explain information in your own words, demonstrating understanding of its meaning.

Example

Explain why word-level tokenization struggles with unknown words.

Explain | Summarize | Describe

Cognitive Domain

APPLY · L3

The ability to apply knowledge in new situations by using what you know to solve problems.

Example

Apply the first three steps of the BPE algorithm to a small corpus.

Implement | Use | Solve

ANALYZE · L4

The ability to break complex information into parts and examine how those parts connect and relate to each other.

Example

Distinguish between the OOV problem in word-level tokenization and the long sequence problem in character-level tokenization.

Compare | Distinguish | Examine

Cognitive Domain

EVALUATE · L5

The ability to make informed judgments using criteria and standards to assess the quality or validity of something.

Example

Evaluate whether removing emojis from review data is a good preprocessing choice for a sentiment analysis task.

Judge | Assess | Argue

CREATE · L6

The highest level of thinking, involving combining elements to create something new, such as a product, plan, or idea.

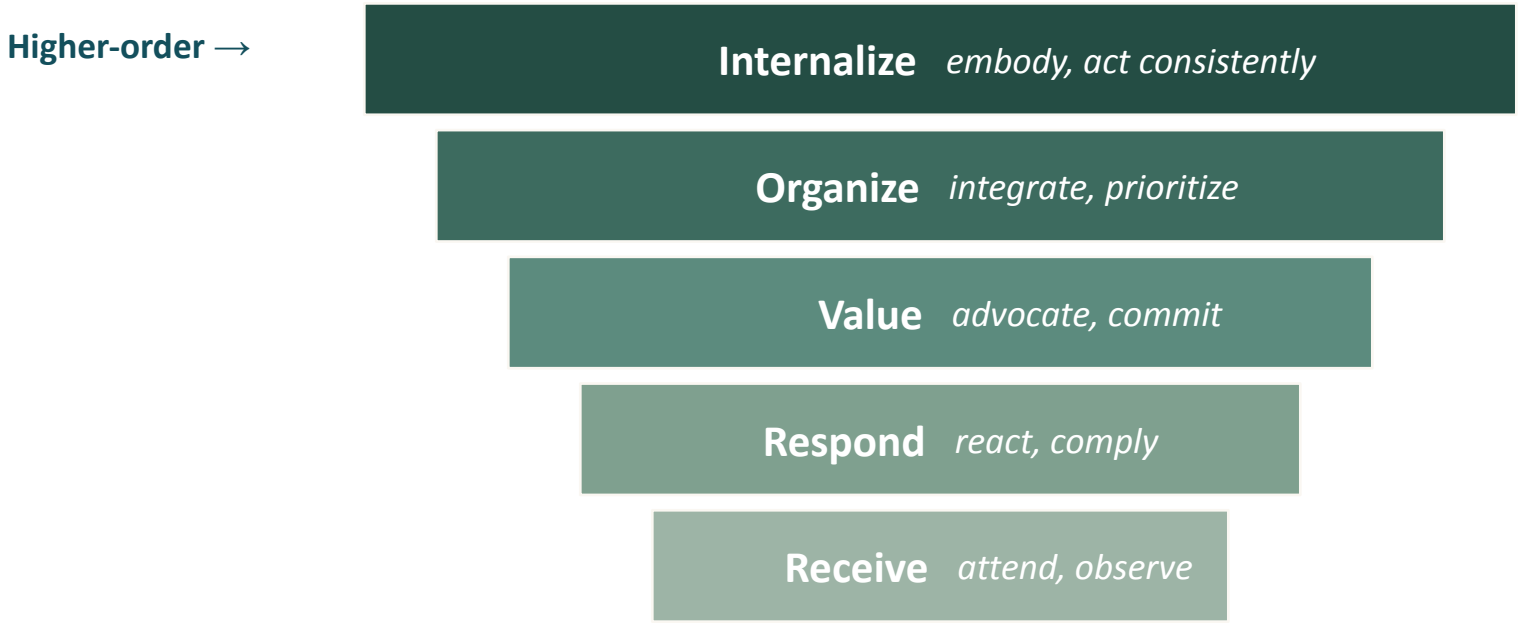
Example

Design a 2D embedding map for ten words from your own language community.

Design | Build | Plan

The Affective Domain

DEFINITION The affective domain is about how learners develop values and ethics, from receiving information to internalizing it as part of their behaviour.



Affective Domain

RECEIVE · L1

Students become aware of an attitude, behaviour, or value and show willingness to pay attention to it.

Example

Listen to a community member describe how they use an AI tool in daily life.

- Attend
- Observe
- Notice

RESPOND · L2

Students respond to a value or attitude and demonstrate active participation.

Example

Discuss how a community is affected when an AI model does not understand their language.

- Discuss
- Contribute
- Share

Affective Domain

VALUE · L3

Students recognise and demonstrate a value through their involvement or commitment, showing that they care about it.

Example

Advocate for community members having a say in how their language and culture are represented in AI.

Advocate | Commit | Justify

ORGANIZE · L4

Students determine priorities and resolve conflicts between different values, developing a personal value system.

Example

Prioritize local languages over convenience when choosing training data for an African AI project.

Prioritize | Integrate | Balance

Affective Domain

INTERNALIZE · L5

Students act consistently with their internalized beliefs, making the value a natural part of their behaviour.

Example

Embody the habit of asking who is affected by your AI tool before asking how well it performs.

Embody | Maintain | Influence

Online vs. Classroom Learning Objectives

ONLINE COURSES

Remember (L1)

List the types of tokenization: sentence-level, word-level, character-level, and subword tokenization.

CLASSROOM SESSIONS

Analyze (L4)

Compare how two tokenizers split a word and identify which one better preserves its meaning.

FATE Foundation

UCL

AIMS | African Institute for Mathematical Sciences SOUTH AFRICA

AI vs RI (Responsible Innovation)

Kenya - Workshop

AI Research Foundations

with support from Google.org

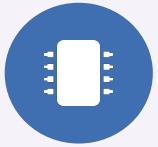
Activities

Two domains, two ladders

COGNITIVE	
Remember	Retrieve
Understand	Explain
Apply	Execute a procedure
Analyze	Tune hyperparameters; compare training curves
Evaluate	Judge competing design choices
Create	Design an MLP architecture

AFFECTIVE	
Receive	Listen / attend
Respond	React / participate
Value	Demonstrate commitment to something
Organize	Integrate different values
Internalize	Integrate values into professional practice

Toolkit activities: AI & RI



AI
Technical skills

- Machine-learning foundations
- Embeddings and neural networks
- Mini-hackathons and end-to-end labs
- Formats: Short - Medium - Long
- Long activities come with lab notebooks



RI
Responsible innovation

- Ethics and responsibility in AI
- Societal impact and fairness
- Reflection and structured discussion
- Formats: Short - Medium
- Facilitation guides

The AI Research Foundations Curriculum

Course

Google DeepMind: 01 Build Your Own Small Language Model

In this Google DeepMind course, you will learn the fundamentals of language models and gain a high-level understanding of the machine...

6 hours

Course Skill badge

Google DeepMind: Train A Small Language Model

Complete the advanced Google DeepMind: Train A Small Language Model skill badge by completing this course to demonstrate skills in...

0%

Course

Google DeepMind: 02 Represent Your Language Data

In this Google DeepMind course you will learn how to prepare text data for language models to process. You will investigate the tools and...

96%

Course

Google DeepMind: 03 Design And Train Neural Networks

In this Google DeepMind course you will focus on the training process for machine learning models. You will learn how to spot and mitigate...

37%

Course

Google DeepMind: 04 Discover The Transformer Architecture

In this Google DeepMind course you will discover the mechanisms of the transformer architecture. You will investigate how...

17%

Course

Google DeepMind: 05 Fine-Tune Your Model

Unleash the power of language models with fine-tuning. In this course, you will learn how to adjust a pre-trained model to a specific task....

94%

UCL Extend: the toolkit's home : extend.ucl.ac.uk



Centre for Engineering Education

AI Research Foundations Toolkit

Tools for teaching responsible AI and machine learning.

Overview

Collapse all

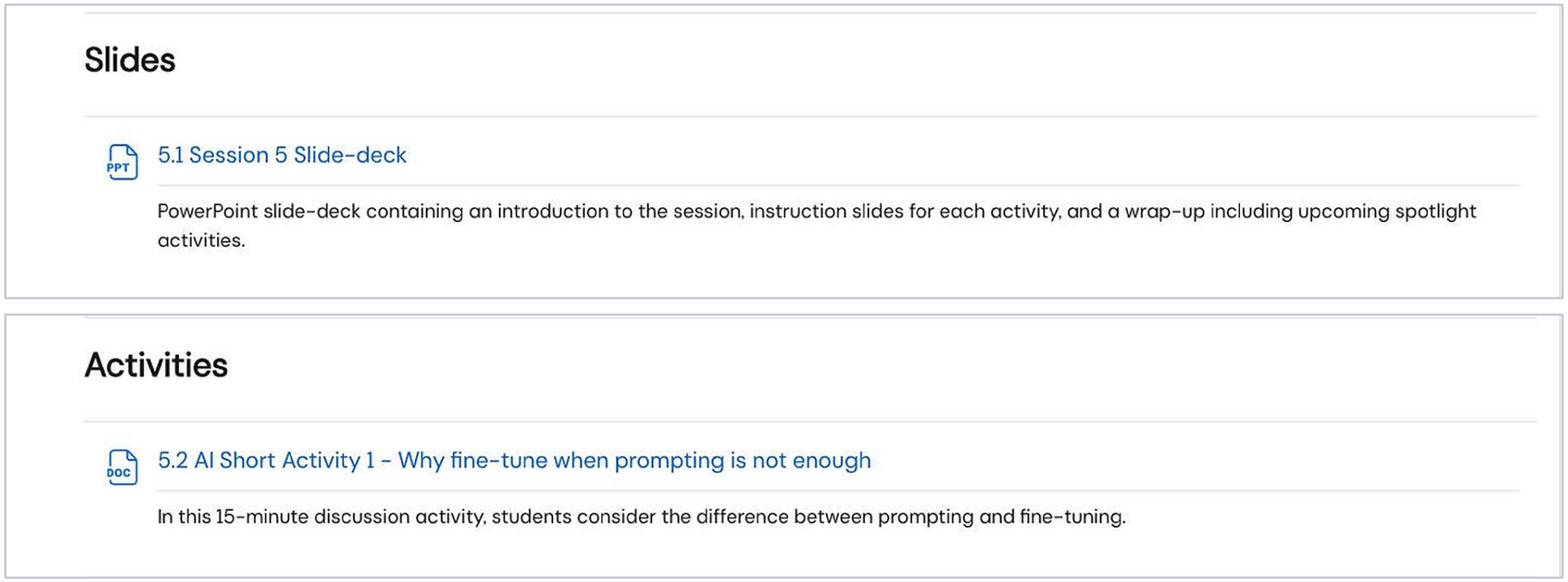
Overview

This open-access toolkit has been developed by University College London (UCL) in collaboration with the African Institute of Mathematical Sciences (AIMS) to widen access to education in responsible AI. Funded by a scientific grant from Google.org, Google's charitable branch, the resources integrate technical learning in AI and machine learning, with responsible innovation (RI) activities that draw on the fields of sociology, philosophy and ethics.

Ten sessions, one page each

- Overview
- Check-in questionnaire
- Google DeepMind: AI Research Foundations
- Session 0
- Session 1
- Session 2
- Session 3
- Session 4
- Session 5
- Session 6
- Session 7
- Session 8
- Session 9
- Assessment
- How to use this toolkit
- Templates and design files
- License

Inside a session page Session 5 shown



RI activities 5.8 to 5.11, further down the same Activities list



Every session page: intro and aims, spotlight pre-learning, slide-deck, ten AI and RI activity files, lab notebooks, learning designs.

One home for everything : UCL Extend

Slides

10 decks

One ready-to-present lecturer deck for every session, 0 through 9.

Learning Activities

10 session pages

AI and Responsible Innovation activities for each session, in Short, Medium and Long formats, wrapped in handbook-style guidance.

Notebooks

21 files

Jupyter labs for the AI Long activities : in Student and Teacher versions plus real datasets.

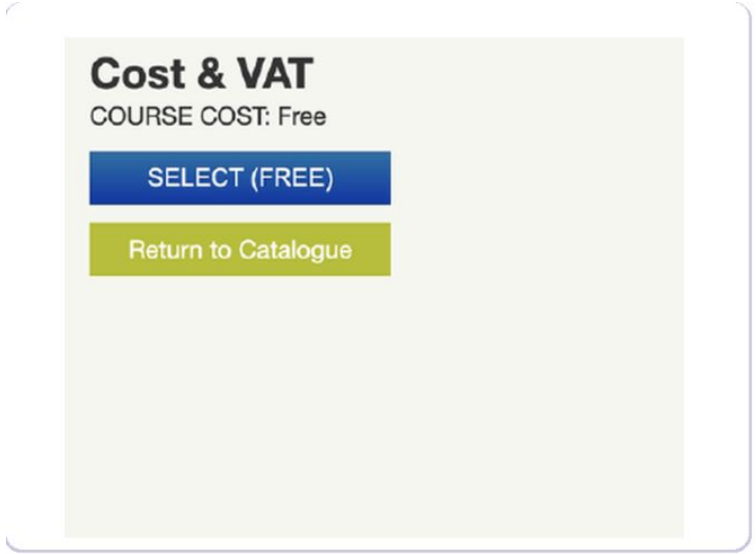
Get the toolkit in five steps

1. **Open the UCL Extend store** (scan the code) and choose **SELECT (FREE)**
2. **Create your account:** choose **“I am a new customer”**, email in lower case
3. **Follow the “Click here” link** under Access Details on the order confirmation page
4. **Open the toolkit** from your UCL Extend dashboard
5. **Complete the check-in questionnaire:** REQUIRED, 3 to 5 minutes; it unlocks the materials

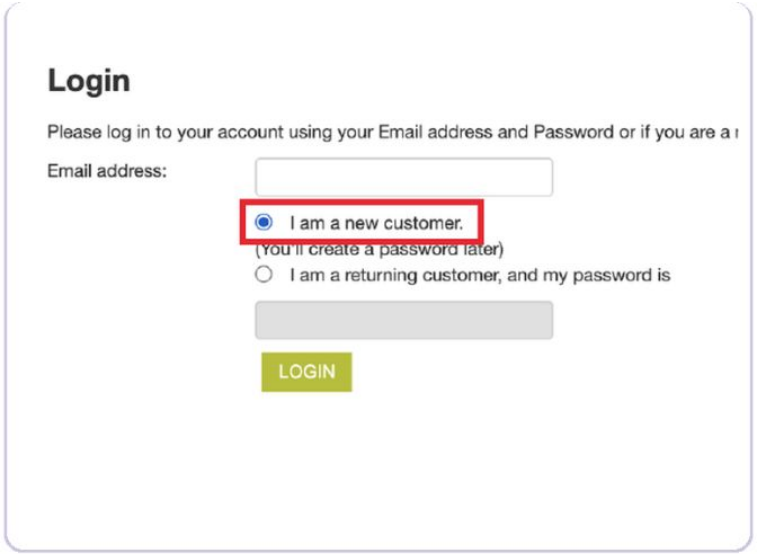
extendstore.ucl.ac.uk

Lecturer Toolkit, free

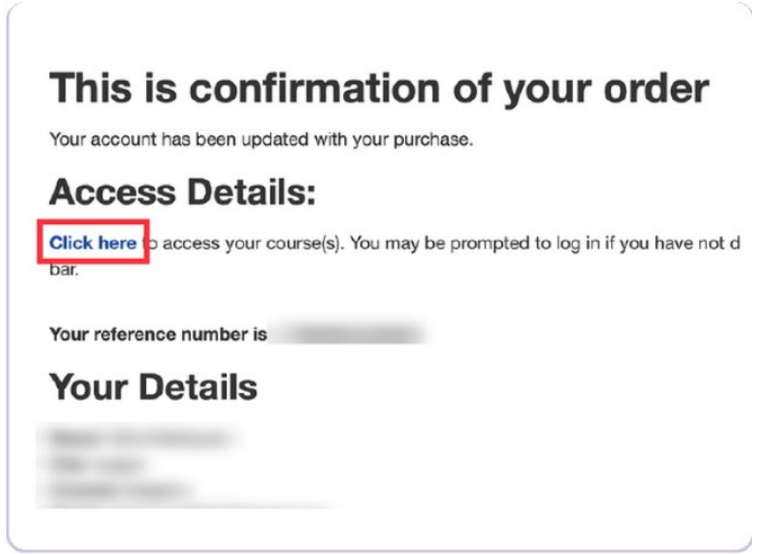
What each step looks like: 1 to 3



1 Choose SELECT (FREE)
extendstore.ucl.ac.uk

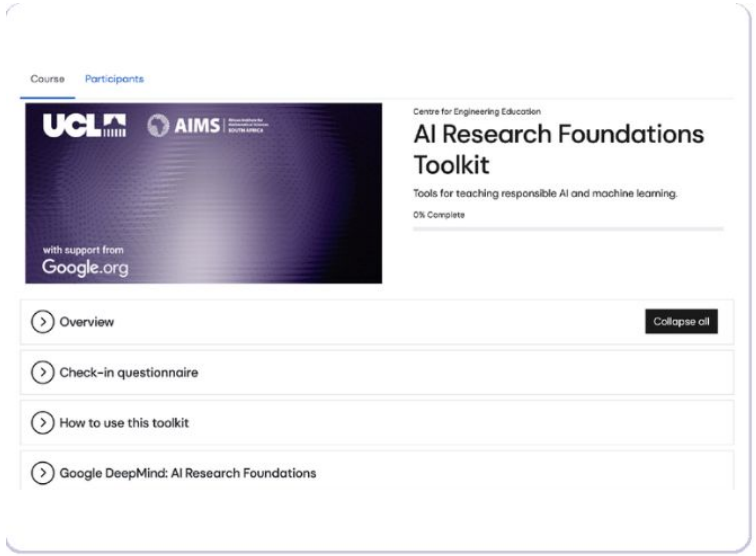


2 I am a new customer
email in lower case

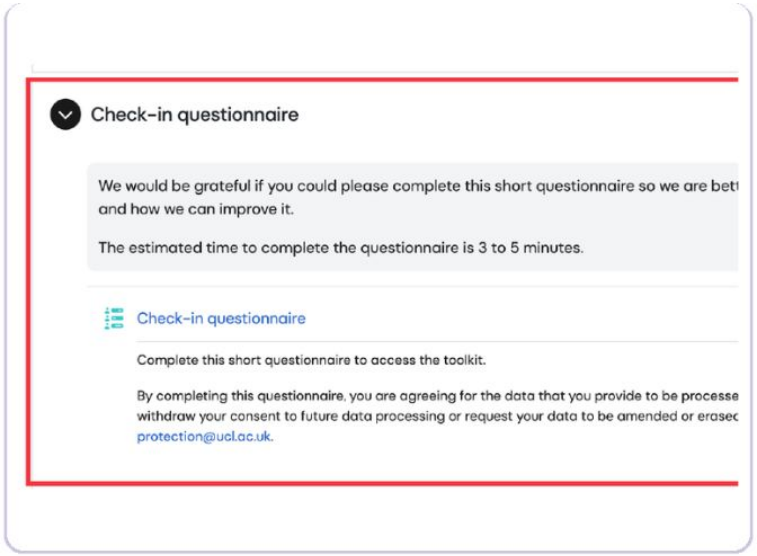


3 Click here (Access Details)
on the order confirmation

What each step looks like: 4 and 5



4 Open the toolkit
from your UCL Extend dashboard



5 Check-in questionnaire
REQUIRED, 3 to 5 minutes

Stuck?

Email us for help

ai-research-foundations @aims.ac.za

Course 2 - Represent Your Language Data

AI Activities

Lab: Implement a BPE Tokenizer

Iteratively merge subword tokens on a small corpus to see how modern tokenizers learn.

RI Activities

Whose voice is missing?

Reflect on the exclusion introduced by dataset selection when training LLMs.

Classify an Activity, AI? RI? Which level?

Example 1

"Train a convolutional neural network to classify maize leaf images as healthy, blighted, or rust-infected using a labelled dataset of 5,000 photographs from Kenyan smallholder farms"

AI | Apply

Example 2

"If your model misdiagnoses a smallholder farmer's only crop, who bears the cost?"

RI | Respond

AI Activity or RI Activity?
Which level in Bloom's Taxonomy?

Question 1 · AI or RI? Which level?

"Implement a BPE tokenizer from scratch and test it on isiZulu text."

A. AI Activity / Apply

B. RI Activity / Apply

C. AI Activity / Respond

D. AI Activity / Create

Question 1 · Answer

“Implement a BPE tokenizer from scratch and test it on isiZulu text.”

A. ✓ AI Activity / Apply

B. RI Activity / Apply

C. AI Activity / Respond

D. AI Activity / Create

Question 2 · AI or RI? Which level?

"Discuss who owns the data when companies scrape African language websites. Use the Fair Trade analogy to propose a governance model."

A. AI Activity / Value

B. RI Activity / Respond

C. RI Activity / Value

D. AI Activity / Analyze

Question 2 · Answer

"Discuss who owns the data when companies scrape African language websites. Use the Fair Trade analogy to propose a governance model."

A. AI Activity / Value

B. ✓ **RI Activity / Respond**

C. RI Activity / Value

D. AI Activity / Analyze

Question 3 · AI or RI? Which level?

"Compare models trained for 10, 400, and 1000 epochs. Identify overfitting and underfitting in the loss curves."

A. AI Activity / Apply

B. AI Activity / Analyze

C. RI Activity / Value

D. AI Activity / Value

Question 3 · Answer

"Compare models trained for 10, 400, and 1000 epochs. Identify overfitting and underfitting in the loss curves."

A. AI Activity / Apply

B. ✓ AI Activity / Analyze

C. RI Activity / Value

D. AI Activity / Value

Question 4 · AI or RI? Which level?

“Estimate the FLOPs needed to fine-tune an SLM. Experiment with bfloat16 to reduce compute cost.”

A. AI Activity / Apply

B. RI Activity / Respond

C. AI Activity / Understand

D. AI Activity / Organize

Question 4 · Answer

“Estimate the FLOPs needed to fine-tune an SLM. Experiment with bfloat16 to reduce compute cost.”

A. ✓ AI Activity / Apply

B. RI Activity / Respond

C. AI Activity / Understand

D. AI Activity / Organize

Question 5 · AI or RI? Which level?

“Read Adebara et al. (2025). Identify three communities missing from a popular NLP benchmark and propose how to include them.”

A. AI Activity / Analyze

B. RI Activity / Respond

C. RI Activity / Organize

D. AI Activity / Organize

Question 5 · Answer

"Read Adebara et al. (2025). Identify three communities missing from a popular NLP benchmark and propose how to include them."

A. AI Activity / Analyze

B. RI Activity / Respond

C. ✓ **RI Activity / Organize**

D. AI Activity / Organize

Question 6 · AI or RI? Which level?

"A Nairobi hospital wants to deploy an AI diagnostic tool trained for 1000 epochs with no validation set. Role-play a meeting between the developer, a nurse, and a patient advocate."

A. AI Activity / Analyze

B. AI Activity / Value

C. RI Activity / Respond

D. RI Activity / Value

Question 6 · Answer

"A Nairobi hospital wants to deploy an AI diagnostic tool trained for 1000 epochs with no validation set. Role-play a meeting between the developer, a nurse, and a patient advocate."

A. AI Activity / Analyze

B. AI Activity / Value

C. RI Activity / Respond

D. ✓ **RI Activity / Value**

Question 7 · AI or RI? Which level?

“Design an MLP architecture for sentiment analysis on Twi product reviews. Justify your hidden layers and activation functions.”

A. AI Activity / Apply

B. AI Activity / Create

C. RI Activity / Analyze

D. AI Activity / Organize

Question 7 · Answer

"Design an MLP architecture for sentiment analysis on Twi product reviews. Justify your hidden layers and activation functions."

A. AI Activity / Apply

B. ✓ AI Activity / Create

C. RI Activity / Analyze

D. AI Activity / Organize

Three takeaways

01

Two domains, both essential.

- AI : Technical
- RI : Ethical

02

Longer activities, higher levels.

- Cognitive : AI
- Affective : RI

03

Protect the pairing.

AI + RI

Questions?